

Arko Datta

PhD Scholar at SeNSE Lab, IIT Madras

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EDUCATION

Indian Institute of Technology, Madras

Ph.D in Computer Science and Engineering

Sensing and Networked Systems Engineering (SeNSE) Lab, guide: Dr. Ayon Chakraborty

Chennai, Tamil Nadu, India

Jul 2025 (Ongoing)

National Institute of Technology, Durgapur

Masters of Technology in Computer Science and Engineering; GPA: 9.60/10.00

Systems Research Lab (SRL), guide: Dr. Sujoy Saha

Durgapur, West Bengal, India

Aug 2023 – May 2025

Kalinga Institute of Industrial Technology

Bachelor of Technology in Computer Science and Engineering; GPA: 9.31/10.00

Bhubaneswar, Odisha, India

Jul 2018 – Jul 2022

M.P. Birla Foundation Higher Secondary School

ISC (Class XII); Score: 92.6%

Kolkata, West Bengal, India

Apr 2018

AWARDS & ACHIEVEMENTS

Institute Gold Medal: For achieving the highest CGPA in the Department of Computer Science and Engineering, in NIT Durgapur. (Dec 2025)

Institute Best Project: Awarded as the best work among all postgraduate students across departments, in NIT Durgapur. Awarded for I-VayuWise: M.Tech. thesis, which provides an end-to-end intelligent air quality monitoring, mitigation and validation system. (Dec 2025)

Travel Award: Awarded student travel award to present our research work at COMSNETS 2025, held at Chancery Pavilion Hotel, Bengaluru. (Jan 2025)

Second Place, Best Student Idea: Received by our team, "Team PolluSense", in the Student Design Contest, ISSS International Conference on Micro, Nano, and Smart Systems (IC-MNSS 2024). Our team comprised of myself and Pabitra Majhi, hailing from Systems Research Lab, NIT Durgapur and guided by Dr. Sujoy Saha. (Jul 2024)

Travel Award: Awarded student travel award to present our research work at ICDCN 2024, held at IIT Madras. (Jan 2024)

PUBLICATIONS

- **ARGOS: Leveraging Visual Priors for Scalable Wireless Navigation in Dynamic Environments**
Arko Datta, Tharaneeshwaran V U, Aravindh Sriram Kumar A G, and Ayon Chakraborty, PERCOM 2026
- **Adaptive Threshold Learning for Robust First-Path Detection in UWB Ranging**
Shrishti Bekal Mudiakal, Akash Kumar Paul, Arko Datta, Ayon Chakraborty, COMSNETS 2026
- **iAirGuard: A Modular IoT Architecture with Dynamic Sampling and TinyML-Based Fault Detection for Air Quality Monitoring**
Molay Mondal, Arko Datta, Prashanth Durgam, Astom Mondal, Khushi Shyam, Khushi Shyam, Subrata Nandi, Sanghita Bhattacharjee and Sujoy Saha, COMSNETS 2026
- **PolluSenseCheck: Cost Effective TinyML-based Air Quality Monitoring System with In-Built Sensor Fault Detection**
Arko Datta, Joyita Chakraborty, Pabitra Majhi, Astom Mondal, Satyaki Roy, Subrata Nandi and Sujoy Saha, COMSNETS 2025
- **Efficient Air Quality Index Prediction on Resource-Constrained Devices using TinyML: Design, Implementation, and Evaluation**
Arko Datta, Aniruddha Pal, Ranbir Marandi, Nilanjan Chattaraj, Subrata Nandi and Sujoy Saha, ICDCN 2024
- **Real-Time Air Quality Predictions for Smart Cities using TinyML**
Arko Datta, Aniruddha Pal, Ranbir Marandi, Nilanjan Chattaraj, Subrata Nandi and Sujoy Saha, ICDCN 2024

PROJECTS

- Generative Signal Maps with Visual Priors** | IIT Madras July 2025 – Present
- A project granted by ANRF ARG to aid scalable wireless sensing deployments in dynamic scenarios, with the aid of visual priors
 - **Publications:** PERCOM '26, COMSNETS '26
- iPolluSense** | NIT Durgapur March 2024 – May 2025
- A project funded by AI4ICPS IIT Kharagpur, to develop intelligent air and noise quality monitoring device that provides personalised insights for the user's pollution exposure, as well as perform self-diagnosis and correction of sensors.
 - **Publications:** COMSNETS '25, COMSNETS '26
- IntAirSense: Intelligent Air Pollution Monitoring** | NIT Durgapur September 2023 – March 2024
- A project funded by West Bengal DST, to develop Development of air pollution monitoring systems which use machine learning to predict a more granular AQI.
 - **Publications:** ICDCN '24 (*2 publications*)

EXPERIENCE

- Indian Institute of Technology Madras** Chennai, Tamil Nadu, India
Teaching Assistant *January 2026 – Present*
- Assisted in delivering lectures for a batch of 50 professional M.Tech. students from notable companies such as Qualcomm, Cert-In and MeITY
 - Enhanced learning experience of students via demos and evaluation materials
- Indian Institute of Technology Madras** Chennai, Tamil Nadu, India
Web Development Assistant *July 2025 – Present*
- Revamped and maintained the website for the Department of Computer Science and Engineering of IIT Madras.
- National Institute of Technology, Durgapur** Durgapur, West Bengal, India
Teaching Assistant *August 2023 – May 2025*
- Assisted in delivering lectures and perform laboratory activities in the Data Structures, Computer Networks, Compiler Design, and Data Science laboratories.
 - Prepared presentations and teaching materials to supplement Data Structures, Data Science, Internet of Things, and Computer Networks lectures.
- Indian Institute of Technology, Madras** Chennai, Tamil Nadu, India
Summer Researcher *May 2024 – July 2024*
- Was involved in a research on Advanced Capacitive Sensing in the Sensing and Networked Systems Engineering (SENSE) lab at IIT Madras, under Dr. Ayon Chakraborty, department of CSE.
- HighRadius Technologies Pvt. Ltd.** Bhubaneswar, Odisha, India
Associate Software Engineer - I *July 2022 – March 2023*
- Developed product configuration migration solutions to reduce configuration migration process by 90%.
 - Developed product configuration solutions to reduce configuration process by 70%.
 - Trained and mentored a batch of hundred interns to develop configuration and migration solutions.
- HighRadius Technologies Pvt. Ltd.** Remote
Full-Time Intern *April 2021 – July 2022*
- Developed product configuration migration solutions to reduce configuration migration process by 90%.
 - Developed product configuration solutions to reduce configuration process by 70%.
- HighRadius Technologies Pvt. Ltd.** Remote
Part-Time Intern *January 2021 – March 2021*
- Received training in the field of Machine Learning and full stack web development using ReactJS and Java.
 - Developed a full stack AI-enabled B2B Order Management System.

SKILLS

Languages: Python, C/C++, Java, JavaScript

Technical Skills: Systems Development, Firmware Development, Computer Vision, Deep Learning

Soft Skills: Leadership, Teamwork, Quick-Learner, Eye for Detail

Hobbies: Piano, Harmonica, Kalimba